1. Equation of a Straight Line

The **equation of a straight line** is a **formula** that helps you draw a straight line on a graph.

A common way to write it is:

$$y = a+bx$$

where:

- **x** = input (like months, cost, time, effort)
- **y** = output (like revenue, leads, or sales)
- **b** = slope (how fast y changes as x increases)
- **a** = intercept (the value of y when x is 0)

Business Example:

If we want to predict **monthly sales (y)** based on **ad spend (x)**, the equation could be:

$$\text{Sales} = 5 \times \text{Ad Spend} + 1000$$

2. Summation (∑)

What is it?

The symbol **∑** (capital Greek letter sigma) just means "**sum of many things**."

If you see:

$$\sum_{i=1}^{5} x_i$$

It means: $x_1 + x_2 + x_3 + x_4 + x_5$

Business Example:

You track customer feedback scores over 5 days:

1	4
2	5
3	3
4	4
5	5

Then:

$$\sum_{i=1}^{5} \text{Score}_i = 4 + 5 + 3 + 4 + 5 = 21$$

This helps when calculating averages or totals.

3. Absolute Value (| |)

What is it?

The **absolute value** of a number is its distance from zero, **always positive**.

$$|-3| = 3, |3| = 3$$

Business Example:

Say we predicted ₹5000 in revenue, but the actual was ₹4500. The error is:

$$\text{Error} = |\text{Actual Value} - \text{Predicted Value}| = |4500 - 5000| = |-500| = 500$$

This is important because we would like to measure how far off our predictions are, regardless of whether we overestimated or underestimated.

Why do the Above Concepts Matter in Linear Regression?

- The **equation of a line** helps us create a model that predicts an output based on an input.
- **Summation (∑)** helps us calculate errors that indicate how good or bad our model is.
- **Absolute value (| |)** ensures we're measuring errors without negative values.